Generalized Linear Mixed Model Results

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1 Introduction

The aim of this study is to investigate factors associated with students' beliefs about whether they will graduate from upper secondary school and obtain the Finnish matriculation examination certificate. The analysis focuses on the relationships between students' responses and selected background characteristics, including age, gender, native language, and year of study.

Since students are grouped by Center for Economic Development, Transport and the Environment (ELY) region, their responses may not be fully independent. Students from the same region may share characteristics that influence their responses. Therefore, a generalized linear mixed model (GLMM) with a binary response and a random intercept for region is used to account for potential differences between regions.

The analysis is conducted in R using a logistic link function. The aim is to examine how the selected background characteristics are associated with the probability that a student believes they will graduate, while accounting for variation between regions. Model adequacy is also assessed using diagnostic plots and statistical tests.

ChatGPT-6 was used to assistance with code development and debugging. It was also used to improve the clarity and grammar of the report. The statistical modelling, interpretation of the results, and final conclusions were reviewed by the author.

2 Data

The data used in this study is from the Finnish Social Science Data Archive (FSD) and is called Upper Secondary School Student Barometer (2024). The dataset contains survey responses from upper secondary school students in Finland. The aim of the analysis is to examine factors associated with students' expectations of graduating from upper secondary school.

The response variable was based on the question "Do you believe that you will graduate from upper secondary school?" The original variable contained several response categories. For this analysis, the variable was recoded as a binary variable. Students who answered "Yes, I believe that I will graduate" were coded as 1, whereas students who selected one of the other substantive response options were coded as 0. Missing or non-substantive responses were excluded from the analysis.

The data also contains information on the ELY center associated with the student's educational institution. The variable ELY is a categorical grouping variable with ELY centers as its categories. It was used as grouping variable in the generalized linear mixed model to account for potential differences between ELY center areas.

Several characteristics were included as explanatory variables. These variables were age, gender, native language and year of studies. The explanatory variables were chosen because they may be associated with students' expectations of graduating. To analyze the data, variables were modified into factors. Gender was treated as a categorical explanatory variable with three categories: female, male, and other. Female students were used as the reference category. Missing responses and responses indicating that the respondent did not wish to disclose their gender were treated as missing. Language was included as a categorical explanatory variable with three categories: Finnish, Swedish, and other. Finnish was used as the reference category. Year of studies was also categorical explanatory variable with categories: first, second, third, fourth or other. Reference group was first year.

In the following figure, we can see portions of students believing in their graduation based on their gender. In figure 1 It is noticed that biggest portion of those, who don't believe in their graduation

belong to the gender "other". As expected, most of the students believe that they will graduate. In figure 2, it looks like the portion of students who believe in their graduation rises until it is "fourth or other" year. In that bar, student graduation expectations decreases seemingly.

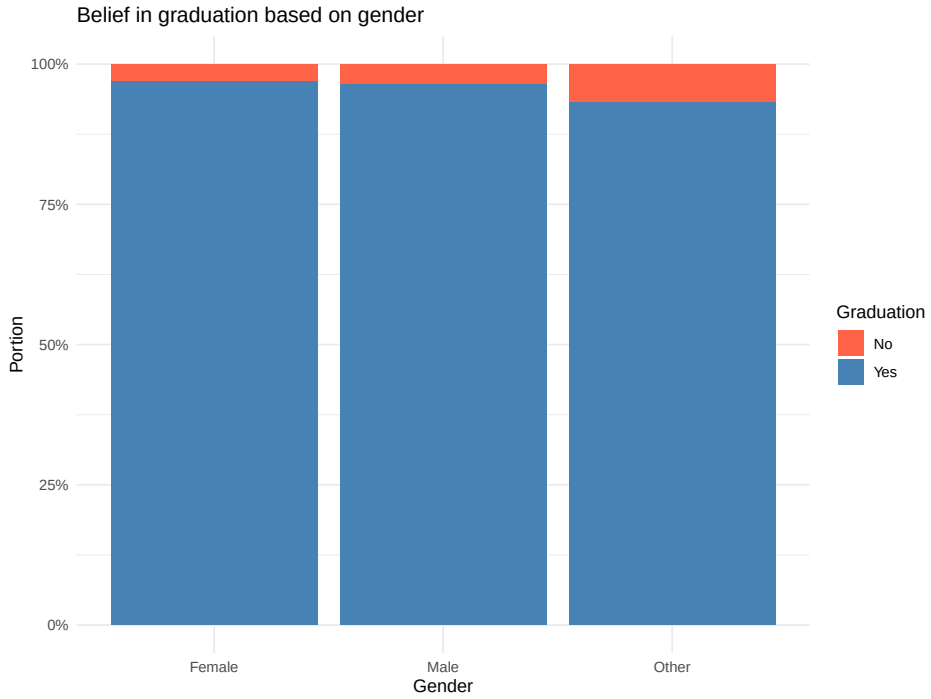


Figure 1: Belief in graduation based on gender.

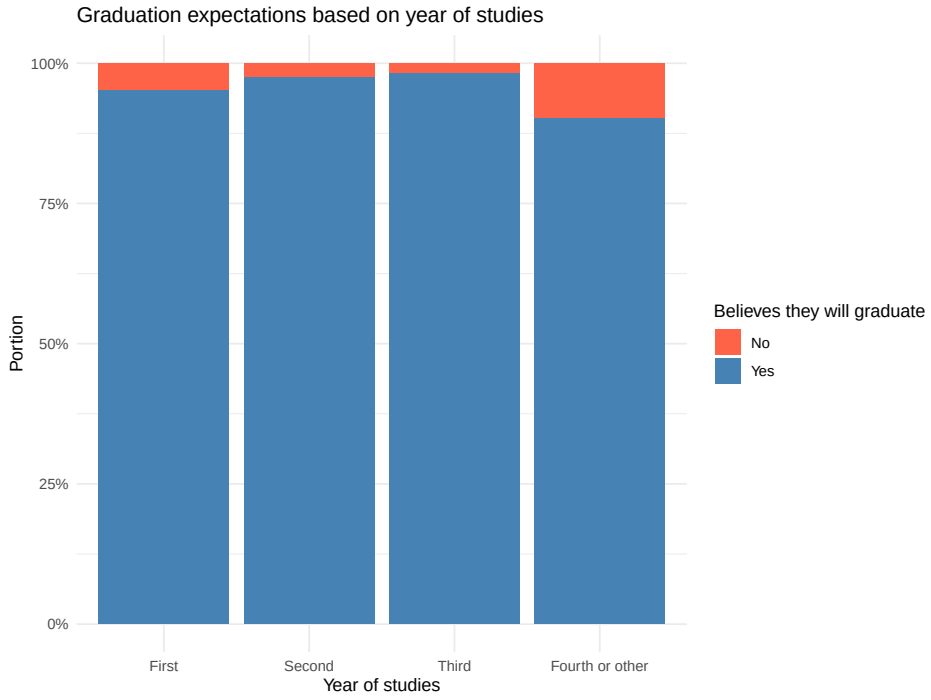


Figure 2: Belief in graduation based on the year of studies.

3 Process

Response variable is binomial and can be expressed as

$$Y_{ij} = \begin{cases} 1, & \text{if the student believes they will graduate,} \\ 0, & \text{otherwise.} \end{cases}$$

We are studying what factors are associated with students' expectations of graduating, and does the association vary between ELY centers. In this project we study factors

1. Age
2. gender [Female, Male, Other]
3. native language [Finnish, Swedish, Other]
4. year of studies. [First, Second, Third, Fourth or other]

At first we model the data using basic logistic regression, which assumes student observations are independent from each other. Logistic model is

$$\text{logit}(P(Y_{ij} = 1)) = \beta_0 + \beta_1 X_{1ij} + \beta_2 X_{2ij} + \beta_3 X_{3ij} + \beta_4 X_{4ij}, \tag{1}$$

where i describes the student, j the ELY center and u_j the random effect of school.

As a second model we use logistic generalized linear mixed model

$$\text{logit}(P(Y_{ij} = 1 \mid u_j)) = \beta_0 + \beta_1 X_{1ij} + \beta_2 X_{2ij} + \beta_3 X_{3ij} + \beta_4 X_{4ij} + u_j. \tag{2}$$

where u_j is the variation between ELY centers. $u_j \sim \mathcal{N}(0, \sigma_u^2)$ As a hypothesis is that there is variation between ELY centers, so the second model is a better fit.

4 Interpretation of results

After comparing AIC:s, GLMM is slightly better fit for the data because the AIC is smaller in that model. We examine the fit of the model, but with binary response variable, we can't use regular linear regressions diagnostics. Therefore we find R-package DHARMa useful.

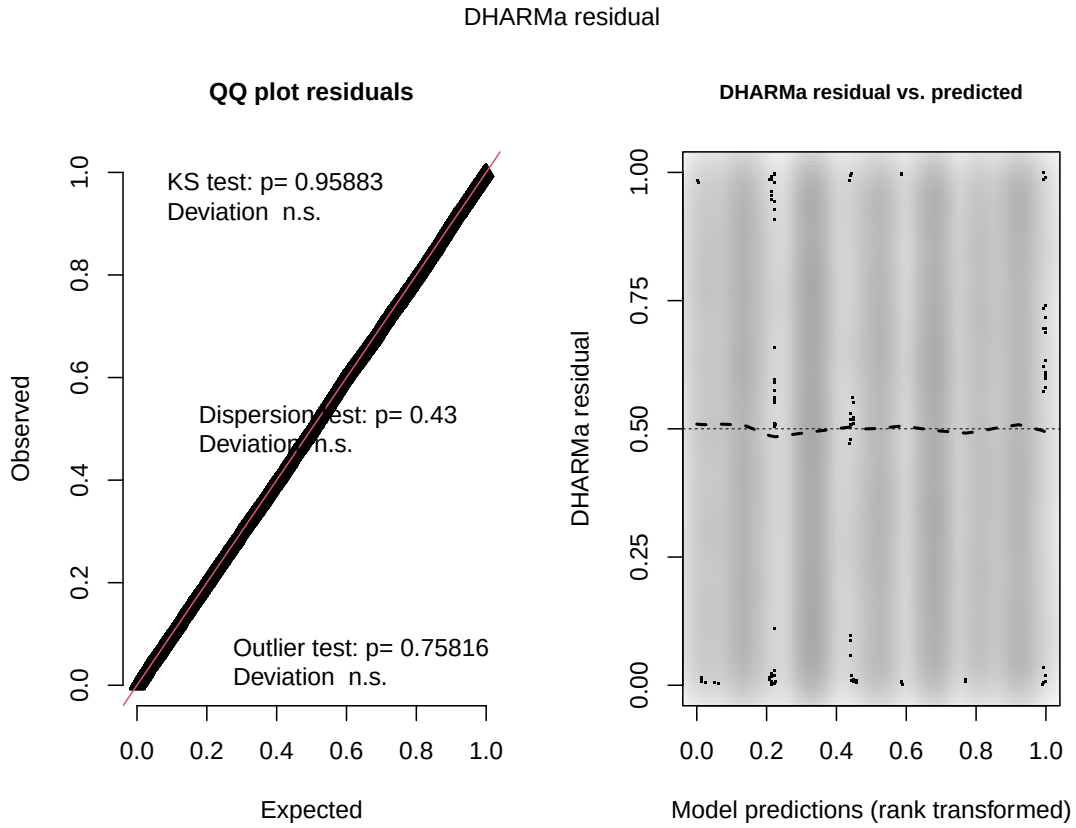


Figure 3: DHARMa residual diagnostic plots for the fitted GLMM.

Model fit was assessed using simulated residuals generated with the DHARMa package. The QQ plot indicated that the residual distribution was consistent with the expected uniform distribution. The Kolmogorov–Smirnov test did not indicate a significant deviation from uniformity (p = 0.959). The dispersion test was also non-significant (p = 0.430), providing no evidence of overdispersion or underdispersion. Furthermore, the outlier test did not detect a significant excess of outliers (p = 0.758).

The residuals versus predicted values plot showed no clear systematic pattern, and the smoothed residual line remained close to 0.5. Overall, these diagnostic results did not indicate major problems with model fit. (Hartig, 2026)

We next interpret the results of the second model. For age, the estimated coefficient is $\hat{\beta}_a = -0.1477$, giving an odds ratio of $\exp(-0.1477) \approx 0.8626$. Thus, a one-year increase in age is associated with approximately 14% decrease in the odds of expecting to graduate, while keeping the other variables constant. However, this is not statistically significant ($p = 0.300$).

Female is used as the reference category for gender. Therefore, the coefficients for male and other describe the differences in graduation expectations compared with females. Being male is associated with approximately 16% lower odds of expecting to graduate. However, this difference is not statistically significant ($p = 0.100$). However being in the other gender category is associated with approximately 51% lower odds, holding the other variables constant is statistically significant.

For native language, Finnish is used as the reference category. Compared with students whose native language is Finnish, students whose native language is Swedish have approximately 44% higher odds of expecting to graduate (not statistically significant), whereas students whose native language is something other than Finnish or Swedish have approximately 50% smaller odds.

First-year students are used as the reference category for year of study. Compared with first-year students, second-year students have approximately 111% higher odds of expecting to graduate, third-year students have approximately 245% higher odds, and students in their fourth year or later have approximately 25% lower odds, holding the other variables constant. Year fourth or other is not

statistically significant.

We find that variance is $\sigma_u^2 \approx 0.0659$ and therefore intraclass correlation is

$$\frac{\sigma_u^2}{\sigma_u^2 + \pi^2/3} = \frac{0.0659}{0.0659 + \pi^2/3} \approx 0.0196. \tag{3}$$

(Nakagawa et al., 2017) This describes how similar observations in the same ELY center are compared to observations in different ELY centers. Consequently, approximately 2% of the variation in the logistic latent scale is attributable to differences between the ELY centers. (Ford, 2026)

5 Conclusion

A generalized linear mixed model (GLMM) with a binomial distribution and logit link was fitted to examine factors associated with students' expectations of graduating. Age, gender, language and year were included as fixed effects, while ELY center was included as a random intercept to account for potential differences between ELY center areas.

Overall, the results of the second GLMM indicate that year of study, gender, and native language are associated with students' expectations of graduating, although not all categories show statistically significant differences. Students in their second and third year had significantly higher odds of expecting to graduate than first-year students. Students in the other gender category and students whose native language was classified as other had significantly lower odds of expecting to graduate compared with their respective reference groups. Age, being male, speaking Swedish, and being in the fourth year or later were not statistically significantly associated with expectations of graduating.

The intraclass correlation coefficient was approximately 0.020, suggesting that only a relatively small proportion of the variation in the latent propensity to expect graduation is attributable to differences between ELY center areas. The GLMM provides a way to examine individual-level factors while accounting for the clustered structure of the data.

6 References

Ford, C. (2026, May 18). *Understanding ICC*. University of Virginia Library. https://library.virginia.edu/data/articles/understanding-icc

Hartig, F. (2026). *DHARMa: Residual diagnostics for hierarchical (multi-level/mixed) regression models*. CRAN. https://cran.r-project.org/web/packages/DHARMa/vignettes/DHARMa.html

Nakagawa, S., Johnson, P. C. D., & Schielzeth, H. (2017). *The coefficient of determination R^2 and intra-class correlation coefficient from generalized linear mixed-effects models revisited and expanded*. Journal of the Royal Society Interface, 14(134), Article 20170213. https://royalsocietypublishing.org/rsif/article/14/134/20170213/116326/The-coefficient-of-determination-R2-and-intra

The data used in this study were obtained from the Finnish Social Science Data Archive (FSD). The dataset is available at: https://services.fsd.tuni.fi/catalogue/FSD3963?tab=summary&study_language=fi.

A R Code

```
    # as binary variable
gra$H_3_bin <- ifelse(gra$H_3 == 1, 1,
                          ifelse(gra$H_3 %in% c(2, 3, 4), 0, NA))
# as categorical variables
gra$A_1 <- factor(
  gra$A_1,
  levels = c(1, 2, 3),
  labels = c("Female", "Male", "Other")
)
gra$A_3 <- factor(
  gra$A_3,
  levels = c(1, 2, 3),
  labels = c("Finnish", "Swedish", "Other")
)

gra$B_1 <- factor(
  gra$B_1,
  levels = c(1, 2, 3, 4),
  labels = c("First", "Second", "Third", "Fourth or other")
)
# bacic logistic regression

model_glm <- glm(H_3_bin ~ A_2 + A_1 + A_3 + B_1, family = binomial, data = gra)
summary(model_glm)
# logistic GLMM
library(lme4)
model_glmm <- glmer(H_3_bin ~ A_2 + A_1 + A_3 + B_1 + (1| OL_ely), family = binomial, data = gra)
summary(model_glmm)

# standard deviation
VarCorr(model_glmm)

# Figure 1
library(ggplot2)
library(dplyr)
gra %>%
  filter(!is.na(A_1), !is.na(H_3_bin)) %>%
ggplot(aes(x = A_1, fill = factor(H_3_bin))) +
  geom_bar(position = "fill") +
  labs(
    x = "Gender",
    y = "Portion",
    fill = "Graduation",
    title = "Belief in graduation based on gender"
  ) +
  scale_y_continuous(labels = scales::percent) +
  scale_fill_manual(
    values = c("0" = "tomato", "1" = "steelblue"),
    labels = c("0" = "No", "1" = "Yes")
  ) +
  theme_minimal()

# Figure 2
library(scales)
gra %>%
```

```
  filter(!is.na(B_1), !is.na(H_3_bin)) %>%
  ggplot(aes(x = B_1, fill = factor(H_3_bin))) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = percent) +
  scale_fill_manual(
    values = c("0" = "tomato", "1" = "steelblue"),
    labels = c("0" = "No", "1" = "Yes"),
    name = "Believes they will graduate"
  ) +
  labs(
    x = "Year of studies",
    y = "Proportion",
    title = "Graduation expectations based on year of studies"
  ) +
  theme_minimal()

# Diagnostics
library(DHARMa)

simulation <- simulateResiduals(
  fittedModel = model_glmm,
  n = 1000,
  simulateREs = "user-specified"
)

plot(simulation)
```

B Results in R

Results of first model

```
Call:
glm(formula = gra_bin ~ age + gender + n_language + year_study, family = binomial, data = data)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)        3.3491     0.1735  19.304  < 2e-16 ***
age               -0.1397     0.1424  -0.981 0.326423
GMale             -0.1892     0.1065  -1.777 0.075580 .
GOther            -0.7112     0.2524  -2.818 0.004830 **
LSwedish           0.2492     0.3630   0.686 0.492402
LOther            -0.7549     0.1731  -4.362 1.29e-05 ***
YSecond            0.7293     0.1839   3.965 7.34e-05 ***
YThird             1.2245     0.3195   3.832 0.000127 ***
YFourthor         -0.2811     0.4732  -0.594 0.552491
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 3456.8  on 11948  degrees of freedom
Residual deviance: 3359.0  on 11940  degrees of freedom
  (1287 observations deleted due to missingness)
AIC: 3377

Number of Fisher Scoring iterations: 6
```

While results of GLMM

```
   Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) [
glmerMod]
 Family: binomial  ( logit )
Formula: gra_bin ~ age + gender + n_language + year_study + (1 | ELY)
   Data: data

      AIC       BIC    logLik -2*log(L)  df.resid
   3357.0    3430.9   -1668.5    3337.0     11917

Scaled residuals:
    Min      1Q  Median      3Q     Max
-9.1948  0.1375  0.1650  0.2006  0.5296

Random effects:
 Groups Name        Variance Std.Dev.
 ELY (Intercept) 0.06586  0.2566
Number of obs: 11927, groups:  ELY, 15

Fixed effects:
                   Estimate Std. Error z value Pr(>|z|)
(Intercept)         3.2939     0.1884  17.484  < 2e-16 ***
age                -0.1477     0.1427  -1.035 0.300718
GMale              -0.1756     0.1068  -1.644 0.100176
GOther             -0.7212     0.2529  -2.852 0.004346 **
LSwedish            0.3630     0.3749   0.968 0.332843
LOther             -0.7033     0.1761  -3.994 6.49e-05 ***
YSecond             0.7455     0.1844   4.044 5.26e-05 ***
YThird              1.2389     0.3200   3.872 0.000108 ***
YFourth or other   -0.2881     0.4738  -0.608 0.543160
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation of Fixed Effects:
          (Intr)  age   GMale  GOth   LSwd   LOth   YScn   YThr
age       -0.798
GMale     -0.221 -0.027
Gther     -0.093  0.001  0.180
LSwedish  -0.058  0.006 -0.012 -0.010
LOther     0.103 -0.263  0.003 -0.015  0.081
YSecond    0.461 -0.752  0.020 -0.011 -0.004  0.221
YThird     0.612 -0.868  0.028 -0.006 -0.014  0.249  0.735
YFourthor  0.606 -0.822  0.035 -0.083 -0.002  0.223  0.674  0.747
```